Fairway Accounting Group Security Overhaul

Technical Challenges and Troubleshooting Report

Optional Supporting Document

1. Purpose

During the Fairway Accounting Group project, I ran into several technical problems while configuring and testing the virtual network. Some issues involved network connectivity, while others involved system configuration and security tools.

This document explains some of the problems I encountered, how I investigated them, and what I learned from resolving them. It is separate from the main security audit report because its purpose is to demonstrate my troubleshooting process rather than repeat the assessment findings.

2. DNS Resolution and Wazuh Agent Installation

Problem: While installing the Wazuh agent on the Windows workstation, the system could not properly resolve the download server's hostname.

Investigation: I checked the workstation's DNS settings and confirmed that it was using the Fairway domain controller as its DNS server. I then tested external name resolution and found that the domain controller was not successfully forwarding external DNS requests.

Resolution: I configured the domain controller's DNS service to forward external requests to 8.8.8.8. After making the change, I repeated the DNS lookup and confirmed that the workstation could resolve the Wazuh download server.

The agent installation then completed successfully, and the workstation appeared in the Wazuh dashboard.

What I learned: A problem that appears to involve an application or installer can actually be caused by an underlying network service. Checking DNS resolution before changing other settings helped identify the actual cause.

3. Network Segmentation and Firewall Troubleshooting

Problem: Moving the client portal into a separate network required changes to the virtual network adapters, pfSense interfaces, IP addressing, and firewall rules.

Investigation: I checked the web server's network configuration, verified its default gateway, and tested connectivity between the original LAN and the new server segment. I also reviewed the pfSense rules to determine which connections should be allowed.

Resolution: I configured the portal on the 10.10.20.0/24 network and used pfSense and the Ubuntu firewall to control access. I then tested HTTPS and SSH connections from different machines to confirm that authorized traffic worked while unnecessary access was restricted.

What I learned: Network segmentation involves more than creating a new subnet. Routing, firewall rules, gateways, and application requirements must all work together.

4. Wazuh Installation and Service Troubleshooting

Problem: Setting up Wazuh required troubleshooting installation and service startup issues. At one point, the Wazuh dashboard could not communicate properly with its backend services.

Investigation: I checked the Wazuh services and reviewed whether the manager, indexer, and dashboard were running. I also verified the server's available storage and network configuration.

Resolution: I expanded the virtual server's available storage during setup and resolved a later indexer startup issue by restarting the affected service. Once the services were operational, I confirmed that the dashboard was accessible and that the Windows agent was reporting.

What I learned: Monitoring systems depend on several connected services. Checking each component individually is more effective than assuming the dashboard itself is the source of the problem.

5. Virtual Machine Startup Issues

Problem: Some Ubuntu virtual machines experienced startup problems and did not always boot normally.

Investigation: I reviewed the startup behavior and used the GRUB boot menu to test whether the problem was related to the virtual machine's boot configuration.

Workaround: I was able to start the affected systems by adjusting the boot parameters, including using the nomodeset option when needed.

What I learned: Virtualization problems can affect lab availability even when the network and security configurations are correct. This issue was worked around for testing but was not permanently resolved.

6. Overall Lessons Learned

The biggest lesson from this project was the importance of troubleshooting problems in a logical order. Instead of changing several settings at once, I learned to identify what was failing, test possible causes, and verify the result after making a change.

I also learned that security tools depend heavily on basic networking services such as DNS, routing, and firewall configuration.

These troubleshooting experiences helped me become more comfortable working with unfamiliar systems and gave me practical examples of problems I can explain during a technical interview.